

Community Library Membership & Book Lending System

A C# .NET Core console application using Entity Framework Core and PostgreSQL to manage library operations.

ER Diagram

[ER Diagram](#)

Table Relationships

- **Membership (1) to Member (Many):**
A membership tier can be assigned to multiple members.
 - **Member (1) to Borrowing (Many):**
A member can borrow multiple books over time.
 - **BookCategory (1) to Book (Many):**
A category can contain multiple books.
 - **Book (1) to BookCopy (Many):**
A single book title can have multiple physical copies.
 - **BookCopy (1) to Borrowing (Many):**
A physical copy can participate in multiple borrowing records over time.
 - **Borrowing (1) to Fine (Many):**
A borrowing record can generate multiple fines such as late return fines or damage fines.
 - **Member (1) to MembershipPayment (Many):**
A member can make multiple membership payments over time.
 - **BookCopy (1) to BookDamageHistory (Many):**
A book copy can have multiple damage reports.
 - **Member (1) to BookDamageHistory (Many):**
A member can be associated with multiple damage incidents.
-

Many-to-Many Relationships

- **Member to BookCopy (Many-to-Many):**

Resolved through the **Borrowing** junction entity.

- One member can borrow many book copies.
 - One book copy can be borrowed by many members over time.
-

History Entity

- **BookDamageHistory:**

Acts as a damage tracking entity that records:

- damaged copy
- responsible member
- severity
- damage description
- damage date

Functional Requirements

Member Functional Requirements

Note: For simplicity, App uses only email ID for login.

- Register as a new member with a name, email, and phone number.
- View personal profile and current active membership status.
- Purchase or upgrade to a membership tier (Basic, Student, Premium).
- Search the book catalog by Title/Author or by Category as separate features.
- View available books that have copies currently in stock.
- Borrow a book copy under active membership limits.
- View currently borrowed books and their due dates.
- Return a borrowed book.
- View and pay pending fines.

Admin Functional Requirements

- Manage memberships by creating, viewing, updating, or deactivating tiers.
- Manage the book catalog by adding books and generating multiple physical book copies with unique copy numbers.
- Mark individual copies as Damaged and charge the member a fine based on severity (Low, Medium, High).
- View active damage history reports.
- Manage categories by adding, listing, updating.
- Manage members by deactivating them.
- Run administrative reports:
 - Books currently checked out.
 - Overdue book copy reports.
 - Members with active unpaid fines.
 - Most frequently borrowed books.
 - Total available books summarized by category.
 - Full borrowing history audit for any member.

Important Logic and Application Outputs

1. Duplicate Borrowing Prevention

A member cannot borrow a duplicate copy of a book they already currently check out.

```
Select an option: 5
--- Borrow a Book ---
Enter Book ID to borrow: 5
Failed to borrow: You have already borrowed this book and have not returned it yet.
```

2. Membership Tier Borrowing Limit Exceeded

Basic members are restricted to 2 active borrowings (7-day duration limit), Student to 3 (10-day duration limit), and Premium to 5 (15-day duration limit).

```
Select an option: 5
--- Borrow a Book ---
```

```
Enter Book ID to borrow: 3
```

```
Failed to borrow: Member has reached the maximum borrow limit.
```

3. Block Borrowing Due to Unpaid Fines Exceeding 500 Rupees

Members with unpaid fines exceeding 500 rupees are blocked from checking out any more books until fines are cleared.

```
Select an option: 5
```

```
--- Borrow a Book ---
```

```
Enter Book ID to borrow: 2
```

```
Failed to borrow: Member has unpaid fines exceeding ₹500. Please pay fines to continue borrowing.
```

4. Late Return Fines

If a book is returned after its due date, the system automatically generates a pending fine calculated at a rate of **₹10 per day late**.

5. Deactivating a Member Blocks Authentication

Deactivating a member soft-deletes their account, making them unable to log in.

```
Select an option: 4
```

```
--- Delete Member ---
```

```
Enter Member ID: 5
```

```
Are you sure you want to deactivate member 5? (y/n): y
```

```
Member deactivated successfully!
```

```
---
```

```
Select an option: 3
```

```
Enter your email: tony@gmail.com
```

```
[Not Found] Member with 'Email : tony@gmail.com' not found.
```

5. Input Validation Failures

Ensures no empty titles or authors can be submitted for cataloging.

```
Select an option: 1
--- Add New Book ---
Title:
Author:
ISBN:
Published Year:
Category ID:

[Validation Error] Book title and author cannot be empty.
```

Exception Handling

The application implements a robust, three-tier exception handling architecture that enforces a clean **Separation of Concerns** across all layers:

1. Separation of Concerns & Exception Propagation Flow

- **Data Access Layer (DALLibrary):** Catches database/EF Core errors and throws a custom `RepositoryException` to shield raw internals.
- **Business Logic Layer (BLLibrary):** Throws `ValidationException` for business rule violations. Catches `RepositoryException` from the DAL and wraps it into a `ServiceException`.
- **Presentation Layer (FELibrary):** Passes bubbled exceptions to `ConsoleHelper.HandleException` to display clear, colorized, user-friendly alerts instead of technical stack traces.

2. Custom Exceptions Reference

- **ValidationException:** Thrown by BLL when validation or borrowing rules fail. Outputs prefix: `[Validation Error]`.
- **DataNotFoundException:** Thrown by BLL when a requested record (Member, Book, or Category) is missing. Outputs prefix: `[Not Found]`.
- **RepositoryException:** Thrown by DAL (`SaveChanges()`) when a raw database query or transactional operation fails.
- **ServiceException:** Wraps BLL service execution failures or caught repository errors. Outputs prefix: `[Service Error]`.

How to Run the Code

1. Database Setup

1. Open PostgreSQL and create a database named `BookRental`.
2. Apply the table schemas and seed values.
3. Register the custom stored function `get_total_unpaid_fine` using the following query:

```
CREATE OR REPLACE FUNCTION get_total_unpaid_fine(member_id integer)
RETURNS numeric AS $$
DECLARE
    v_total numeric;
BEGIN
    SELECT COALESCE(SUM(f."FineAmount"), 0)
    INTO v_total
    FROM "Fines" f
    JOIN "Borrowings" b ON f."BorrowingId" = b."BorrowingId"
    WHERE b."MemberId" = member_id AND f."PaidStatus" = 1;

    RETURN v_total;
END;
$$ LANGUAGE plpgsql;
```

2. Configure Connection String

Ensure the connection string in `DALLibrary/Contexts/BookRentalContext.cs` is configured with your correct local database credentials (e.g., host, port, username, password).

3. Build and Run the Console Application

Navigate to the root workspace folder in your terminal and run the following commands:

```
dotnet build
dotnet run --project FELibrary
```

Application Output Screenshots

Here are the output screenshots showing the complete console application interface and operational scenarios:

1. Main Menus

Login Menu

```
=== Login Menu ===  
1. Admin operation  
2. Register  
3. Login  
0. Exit
```

```
Select an option: █
```

Login Menu

Admin Menu

```
=== Admin Menu ===  
1. Membership Management  
2. Book Management  
3. Reports  
4. Member Management  
5. Category Management  
0. Logout
```

```
Select an option: █
```

Admin Menu

2. Member Operations

Register Member

```
--- Register New Member ---  
Enter Name: ram  
Enter Email: ram@gmail.com  
Enter Phone Number: 9342129492  
  
member registered successfully!  
  
press any key to continue...
```

Register Member

Member Profile

```
--- My Profile ---  
ID: 13  
Name: ram  
Email: ram@gmail.com  
Phone: 9342129492  
Membership Status: NOT_PURCHASED  
  
Press any key to continue...
```

Member Profile

Purchase Membership


```
--- Manage Membership ---
1. View My Membership
2. Purchase/Upgrade Membership
0. Back

Select an option: 2

--- Available Memberships ---
ID: 1 | Name: basic | Price: ₹600.00 | Limit: 2 books | Duration: 7 days limit
ID: 2 | Name: student | Price: ₹300.00 | Limit: 3 books | Duration: 10 days limit
ID: 3 | Name: premium | Price: ₹1000.00 | Limit: 5 books | Duration: 15 days limit
ID: 5 | Name: summerplan | Price: ₹160.00 | Limit: 2 books | Duration: 4 days limit

Enter Membership ID to purchase/upgrade: 1
Membership purchased successfully and payment recorded!

Press any key to continue...
```

Purchase Membership

Available Books

```
--- Available Books ---
ID: 5 | Title: BreakingBad | Author: vince gillian
ID: 3 | Title: Sherlock Holmes | Author: Arthur Conan Doyle
ID: 7 | Title: mission imposible | Author: christopher

Press any key to continue...
█
```

Available Books

Borrow Book

```
--- Borrow a Book ---
Enter Book ID to borrow: 5
Successfully borrowed the book!

Press any key to continue...
█
```

Borrow Book

Active Borrowings

```
--- My Borrowed Books ---  
  
Borrowing ID: 13 | Copy ID: 10 |  
Book: 'BreakingBad' by vince gillian|  
Borrowed: 19/05/2026 | Due: 26/05/2026 |  
Status: ACTIVE  
  
Borrowing ID: 14 | Copy ID: 4 |  
Book: 'Sherlock Holmes' by Arthur Conan Doyle|  
Borrowed: 19/05/2026 | Due: 26/05/2026 |  
Status: ACTIVE  
  
Press any key to continue...  
█
```

Active Borrowings

Return Book

```
--- Return a Book ---  
Enter Copy ID to return: 10  
Successfully returned the book!  
  
Press any key to continue...  
█
```

Return Book

3. Admin Catalog & Tiers Control

Book Management

```

=== Book Management ===
1. Add Book
2. Add Copies to Book
3. View All Books
4. View a Book's copies
5. Mark Copy as Damaged
6. Mark Copy as Available
7. View Damage History
0. Back to Admin Menu

Select an option: █

```

Book Management

Add Book Copies

```

--- Add Book Copies ---
Enter Book ID: 8
Enter number of copies to add: 20
20 copies added successfully.

Press any key to continue...
█

```

Add Book Copies

Book Copies Listing

```

Enter the Book id to view its copies
5
15
CopyId: 9, BookId: 5, CopyNumber: d32612a2, Status: AVAILABLE, AddedDate: 18/05/2026 10:00:25 AM
CopyId: 10, BookId: 5, CopyNumber: 579b399b, Status: AVAILABLE, AddedDate: 18/05/2026 10:00:25 AM
CopyId: 11, BookId: 5, CopyNumber: c00db5b0, Status: AVAILABLE, AddedDate: 18/05/2026 10:00:25 AM
CopyId: 12, BookId: 5, CopyNumber: 157827a9, Status: AVAILABLE, AddedDate: 18/05/2026 10:00:25 AM
CopyId: 13, BookId: 5, CopyNumber: cf233c47, Status: AVAILABLE, AddedDate: 18/05/2026 10:00:25 AM
CopyId: 14, BookId: 5, CopyNumber: bc51794a, Status: AVAILABLE, AddedDate: 18/05/2026 10:00:25 AM
CopyId: 15, BookId: 5, CopyNumber: 35acd171, Status: AVAILABLE, AddedDate: 18/05/2026 10:00:25 AM
CopyId: 16, BookId: 5, CopyNumber: 8670190e, Status: AVAILABLE, AddedDate: 18/05/2026 10:00:25 AM
CopyId: 17, BookId: 5, CopyNumber: 75533119, Status: AVAILABLE, AddedDate: 18/05/2026 10:00:25 AM
CopyId: 18, BookId: 5, CopyNumber: 87cac418, Status: AVAILABLE, AddedDate: 18/05/2026 10:00:25 AM
CopyId: 19, BookId: 5, CopyNumber: 86240ee7, Status: AVAILABLE, AddedDate: 18/05/2026 10:00:25 AM
CopyId: 20, BookId: 5, CopyNumber: 8ac73a86, Status: AVAILABLE, AddedDate: 18/05/2026 10:00:25 AM
CopyId: 6, BookId: 5, CopyNumber: 591785d4, Status: DAMAGED, AddedDate: 18/05/2026 10:00:25 AM
CopyId: 7, BookId: 5, CopyNumber: 6a9fe68f, Status: DAMAGED, AddedDate: 18/05/2026 10:00:25 AM
CopyId: 8, BookId: 5, CopyNumber: 8751121c, Status: DAMAGED, AddedDate: 18/05/2026 10:00:25 AM

```

Membership Menu

```
=== Membership Management ===
1. Create Membership
2. View All Memberships
3. Update Membership
4. Deactivate Membership
5. Activate Membership
0. Back to Admin Menu

Select an option: █
```

Membership Menu

Create Membership

```
--- Create Membership ---
Name: studenttrial
Price: 0
Borrow Limit: 3
Borrow Days Limit: 5
Membership created successfully.

Press any key to continue...
█
```

Create Membership

Memberships List

```
--- View All Memberships ---
ID: 1, Name: basic, Price: 600.00, Borrow Limit: 2, Borrow Days Limit: 7, Status:Active
ID: 2, Name: student, Price: 300.00, Borrow Limit: 3, Borrow Days Limit: 10, Status:Active
ID: 3, Name: premium, Price: 1000.00, Borrow Limit: 5, Borrow Days Limit: 15, Status:Active
ID: 5, Name: summerplan, Price: 160.00, Borrow Limit: 2, Borrow Days Limit: 4, Status:Active
ID: 4, Name: freetrial, Price: 0.00, Borrow Limit: 3, Borrow Days Limit: 4, Status:Inactive
ID: 6, Name: studenttrial, Price: 0, Borrow Limit: 3, Borrow Days Limit: 5, Status:Active

Press any key to continue...
█
```

Mark Book Damaged

```
--- Mark Book Copy as Damaged ---  
Enter Book Copy ID: 10  
Enter User ID to penalize: 13  
Enter Damage Description: torned index page  
Enter Severity (0: LOW, 1: MEDIUM, 2: HIGH): 1  
Book copy marked as damaged, and fine generated successfully.  
  
Press any key to continue...  
█
```

Mark Book Damaged

Book Damage History

```
--- View Damage History ---  
Enter Book Copy ID: 2  
Damage ID: 1 | User: 1 | Severity: HIGH | Desc: torned cover | Date: 18/05/2026 12:16:57 AM  
  
Press any key to continue...  
█
```

Book Damage History

4. Policy Constraint Warnings

Duplicate Borrowing Blocked

```
--- Borrow a Book ---  
Enter Book ID to borrow: 5  
Failed to borrow: You have already borrowed this book and have not returned it yet.  
  
Press any key to continue...  
█
```

Duplicate Borrowing Blocked

Borrow Limit Exceeded

```
--- Borrow a Book ---  
Enter Book ID to borrow: 2  
Failed to borrow: Member has reached the maximum borrow limit.  
  
Press any key to continue...  
█
```

Borrow Limit Exceeded

Fine Payment Blocked

```
--- Fine Management ---  
1. View Pending Fines & Pay  
2. View Fine History  
0. Back  
  
Select an option: 1  
Fine ID: 8 | Amount: ₹300 | Reason: DAMAGED_BOOK | Status: PENDING  
  
Enter Fine ID to pay: 8  
Fine paid successfully!  
  
Press any key to continue...  
█
```

Fine Payment Blocked

Late Return Fine

```
--- Fine Management ---  
1. View Pending Fines & Pay  
2. View Fine History  
0. Back  
  
Select an option: 1  
Fine ID: 7 | Amount: ₹80.00 | Reason: LATE_RETURN | Status: PENDING  
  
Enter Fine ID to pay: 7  
Fine paid successfully!  
  
Press any key to continue...  
█
```

Late Return Fine

5. Admin Reports

Books Currently Borrowed

```
Borrowing ID: 12
Member Name: john
Book Title: Dune
Borrow Date: 2026-05-04
-----
Borrowing ID: 9
Member Name: vikram
Book Title: Dune
Borrow Date: 2026-05-18
-----
Borrowing ID: 14
Member Name: ram
Book Title: Sherlock Holmes
Borrow Date: 2026-05-19
-----
Press any key to continue...
```

Books Currently Borrowed

Overdue Books

```
--- Overdue Books ---
Borrowing ID: 5
Member Name: Bob Jones
Book Title: Harry Potter and the Sorcerer's Stone
Due Date: 2026-05-15
-----
Borrowing ID: 12
Member Name: john
Book Title: Dune
Due Date: 2026-05-11
-----
Press any key to continue...
```

Overdue Books

Members with Pending Fines

```
--- Members With Pending Fines ---
```

```
Member ID: 1
```

```
Name: Alice Smith
```

```
Email: alice@example.com
```

```
Total Fine: $1000.00
```

```
-----  
Member ID: 2
```

```
Name: Bob Jones
```

```
Email: bob@example.com
```

```
Total Fine: $1200.00
```

```
-----  
Press any key to continue...
```

```
█
```

Members with Pending Fines

Member Borrowing History

```
--- Member Borrowing History ---
```

```
Enter Member ID: 11
```

```
Borrowing ID: 12 | Copy ID: 3 |
```

```
Book: 'Dune' by Frank Herbert|
```

```
Borrowed: 04/05/2026 | Due: 11/05/2026 |
```

```
Status: ACTIVE
```

```
-----  
Borrowing ID: 11 | Copy ID: 9 |
```

```
Book: 'BreakingBad' by vince gillian|
```

```
Borrowed: 04/05/2026 | Due: 11/05/2026 |
```

```
Status: RETURNED
```

```
-----  
Press any key to continue...
```

Member Borrowing History

Verification Logs

For further outputs check [Terminal_output.txt](#).